STATA Tutorial 4A

Data Visualization

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Outline

Common Plot Types

Plot Placement

Customizing Appearance

Save and Export Graph

Common Plot Types

- Recall our knowledge on descriptive statistics
- For describing a single variable, what are proper plot types?
  - Continuous variable:
  - Discrete variable:
- For describing bivariate relationships, what are proper plot types?
  - Discrete + Continuous:
  - Continuous + Continuous:
  - Discrete + Discrete:

Common Plot Types: One Variable

One variable

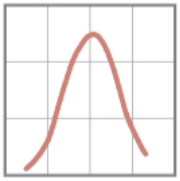
sysuse auto, clear

Continuous

histogram mpg, **width**(5) **freq** **kdensity** **kdenopts(bwidth(5))**

histogram

bin(#) • width(#) • density • fraction • frequency • percent • addlabels
addlabopts(<options>) • normal • normopts(<options>) • kdensity
kdenopts(<options>)



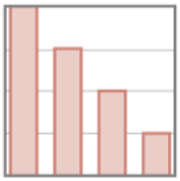
kdensity mpg, **bwidth**(3)

smoothed histogram

bwidth • kernel(<options>
normal • normopts(<line options>)

← main plot-specific options; see help for complete set

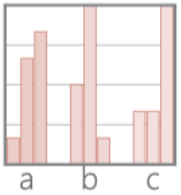
Discrete



graph bar (count), **over**(foreign, **gap**(*0.5)) **intensity**(*0.5)

bar plot — **graph hbar** draws horizontal bar charts

(asis) • (percent) • (count) • over(<variable>, <options: gap(*#) •
relabel • descending • reverse>) • cw •missing • nofill • allcategories •
percentages • stack • bargap(#) • intensity(*#) • yalternate • xalternate



graph bar (percent), **over**(rep78) **over**(foreign)

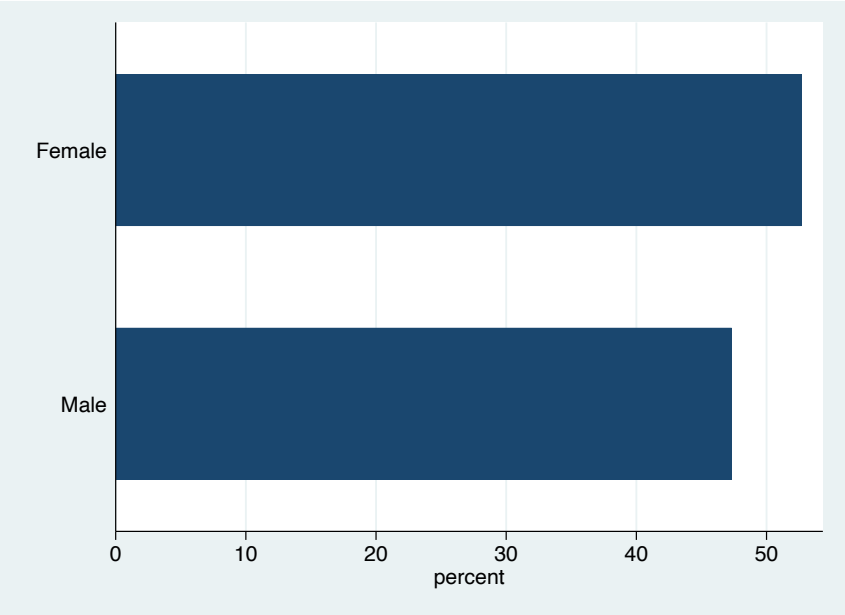
grouped bar plot — **graph hbar** ...

(asis) • (percent) • (count) • over(<variable>, <options: gap(*#) •
relabel • descending • reverse>) • cw •missing • nofill • allcategories •
percentages • stack • bargap(#) • intensity(*#) • yalternate • xalternate

stata cheatsheet

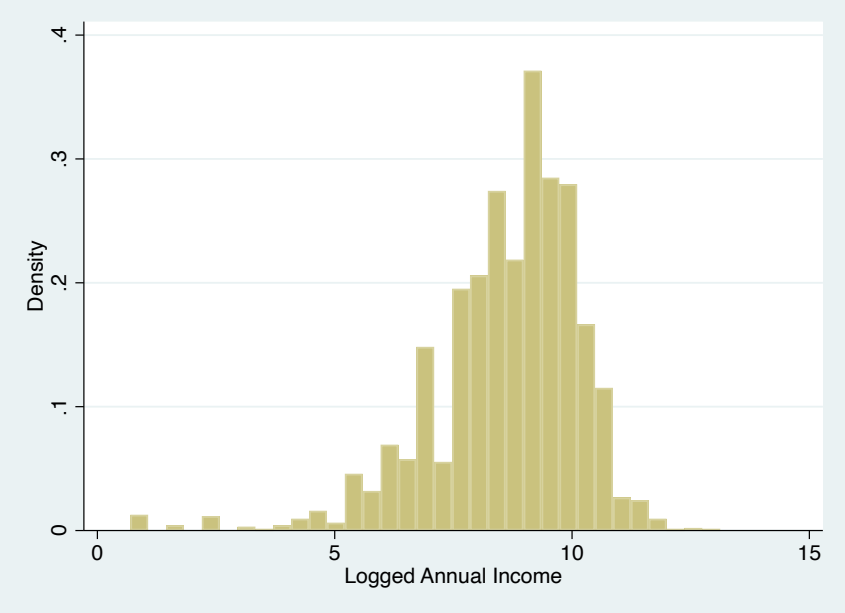
Bar Plot and Pie Chart

```
. * bar
. graph bar (count), over(gender)
. graph hbar (count), over(gender)
. graph bar (percent), over(gender)
. graph hbar (percent), over(gender)
.
. * pie
. graph pie, over(gender)
```



Boxplot, Histogram, and Density Curve

```
. * boxplot
. graph box income
. graph box loginc
.
. * histogram
. histogram income
(bin=35, start=0, width=14285.714)
. histogram loginc
(bin=33, start=.69314718, width=.376
.
. * density curve
. kdensity loginc
```

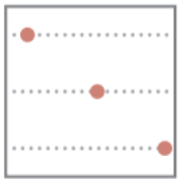


Common Plot Types: Bivariate

Discrete X, continuous Y

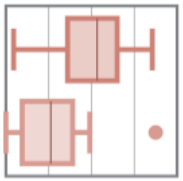
graph bar (median) price, **over**(foreign) — **graph hbar** ...

bar plot (asis) • (percent) • (count) • (stat: mean median sum min max ...)
over(<variable>, <options: gap(*#) • relabel • descending • reverse
sort(<variable>)>) • cw • missing • nofill • allcategories • percentages
stack • bargap(#) • intensity(*#) • yalternate • xalternate



graph dot (mean) length headroom, **over**(foreign) **m**(1, ms(S))

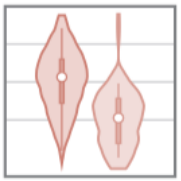
dot plot (asis) • (percent) • (count) • (stat: mean median sum min max ...)
over(<variable>, <options: gap(*#) • relabel • descending • reverse
sort(<variable>)>) • cw • missing • nofill • allcategories • percentages
linegap(#) • marker(#, <options>) • linetype(dot | line | rectangle)
dots(<options>) • lines(<options>) • rectangles(<options>) • rwidth



graph hbox mpg, **over**(rep78, descending) **by**(foreign) **missing**

box plot — **graph box** draws vertical boxplots

over(<variable>, <options: total • gap(*#) • relabel • descending • reverse
sort(<variable>)>) • missing • allcategories • intensity(*#) • boxgap(#)
medtype(line | line | marker) • medline(<options>) • medmarker(<options>)

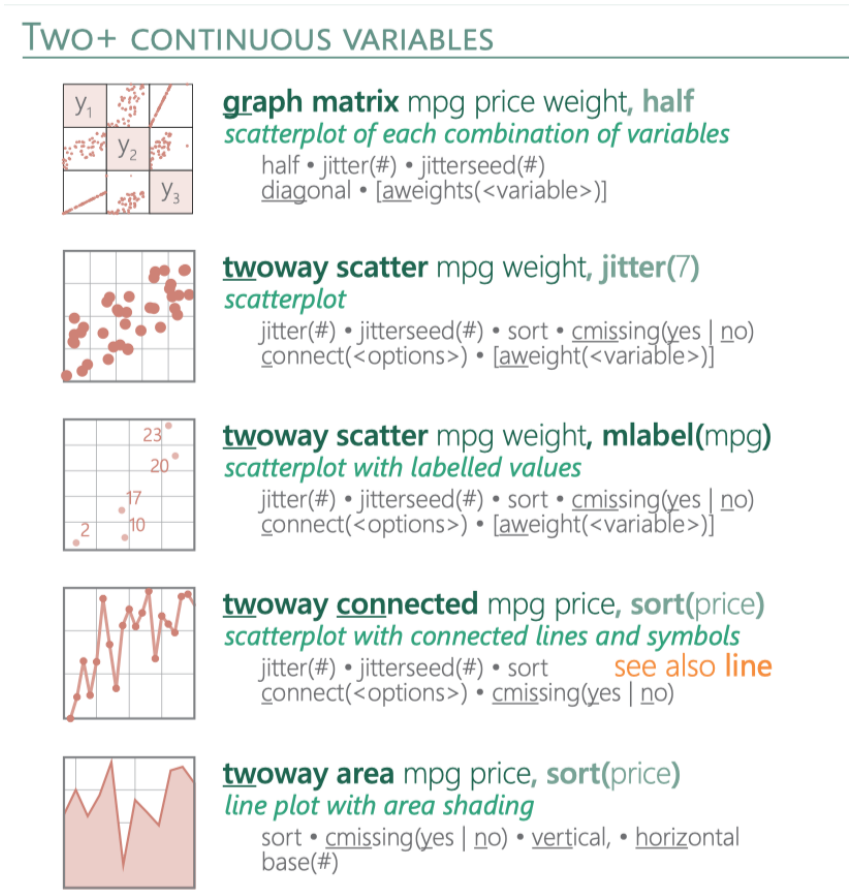


vioplot price, **over**(foreign) — ssc install vioplot

violin plot over(<variable>, <options: total • missing>) • nofill •
vertical • horizontal • obs • kernel(<options>) • bwidth(#) •
barwidth(#) • dscale(#) • ygap(#) • ogap(#) • density(<options>)
bar(<options>) • median(<options>) • obsopts(<options>)

stata cheatsheet

Common Plot Types: Bivariate

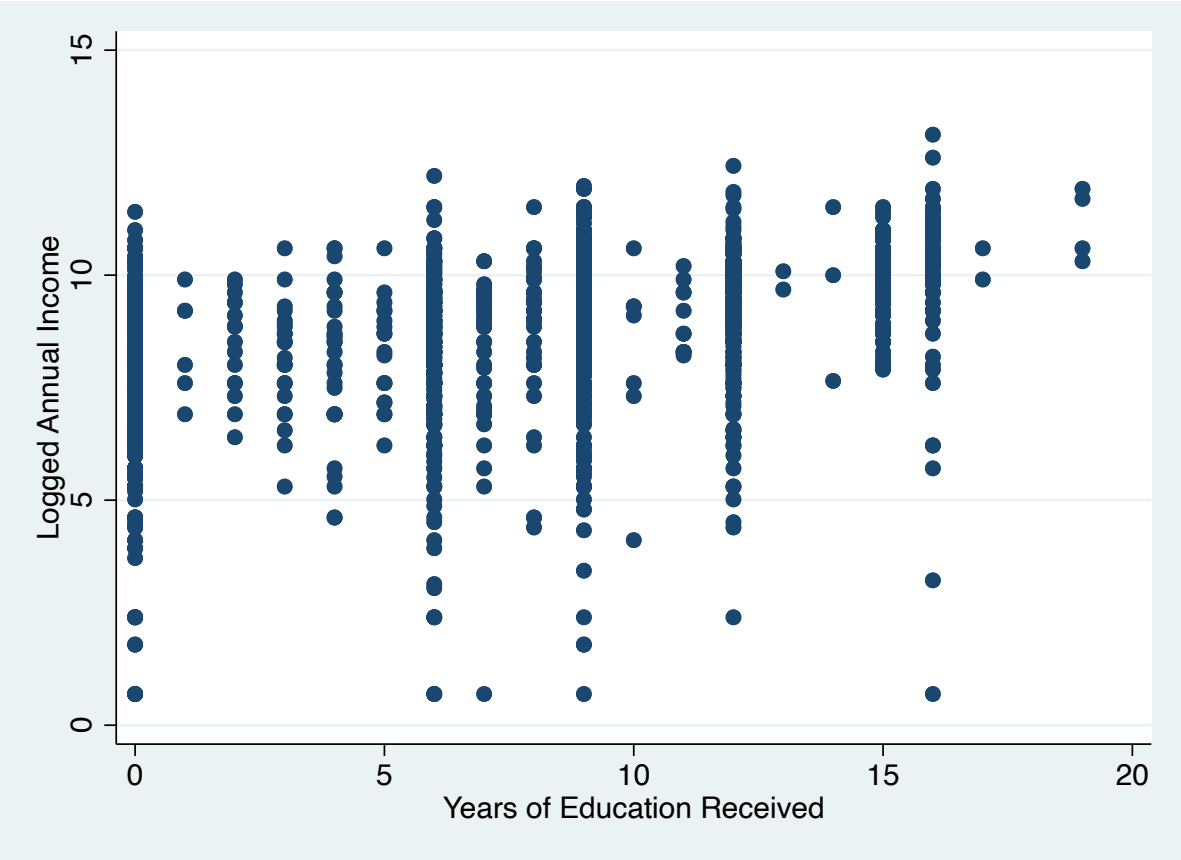


stata cheatsheet

Bivariate Relationships

```
. * scatter plot
. twoway scatter loginc eduy
.
. *** Two Categorical Variables
. graph bar, percentages over(edu) over(urban) asyvars stack
.
. *** Categorical + Continuous
. histogram loginc, by(gender)
. graph box loginc, by(gender)
. graph box loginc, over(gender)
```

Scatter Plot



Outline

Common Plot Types

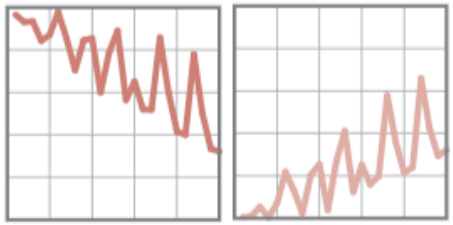
Plot Placement

Customizing Appearance

Save and Export Graph

Plot Placement: Juxtapose vs Superimpose

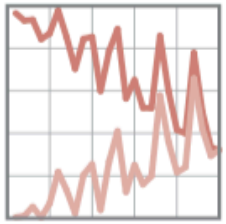
Juxtapose (facet)



twoway scatter mpg price, by(foreign, norescale)

total • missing • colfirst • rows(#) • cols(#) • holes(<numlist>)
compact • [no]edgelabel • [no]rescale • [no]yrescal • [no]xrescale
[no]iyaxes • [no]ixaxes • [no]iytick • [no]ixtick • [no]iylabel
[no]ixlabel • [no]iytitle • [no]ixtitle • imargin(<options>)

Superimpose



graph combine plot1.gph plot2.gph...
combine two or more saved graphs into a single plot

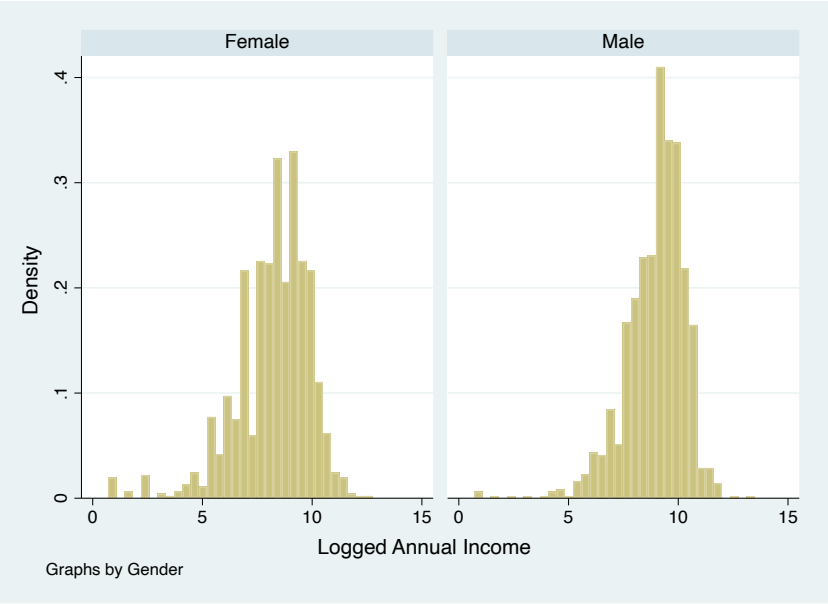
scatter y3 y2 y1 x, msymbol(i o i) mlabel(var3 var2 var1)
plot several y values for a single x value

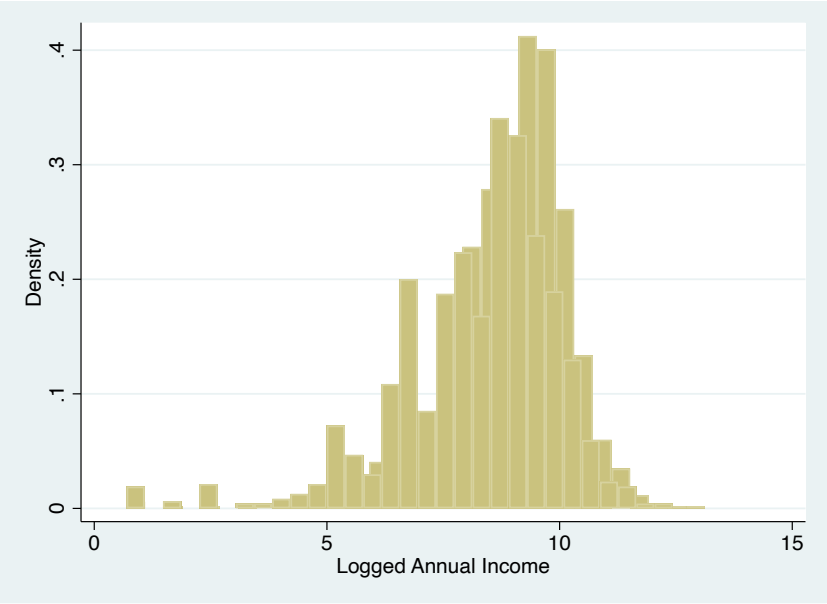
graph twoway scatter mpg price in 27/74 || scatter mpg price /* */ if mpg < 15 & price > 12000 in 27/74, mlabel(make) m(i)
combine twoway plots using ||

Juxtapose vs Superimpose

```
. *** Juxtapose
. histogram loginc, by(gender)
. graph box loginc, over(gender)
. * check the help document for proper options
.
. *** Superimpose
. * Method 1: ||
. twoway histogram loginc if gender==1 || histogram loginc if gender==0
. * Method 2: ()
. twoway (histogram loginc if gender==1) (histogram loginc if gender==0)
```

Juxtapose vs Superimpose





The superimposed plot is somehow hard to read, at least for now.

Outline

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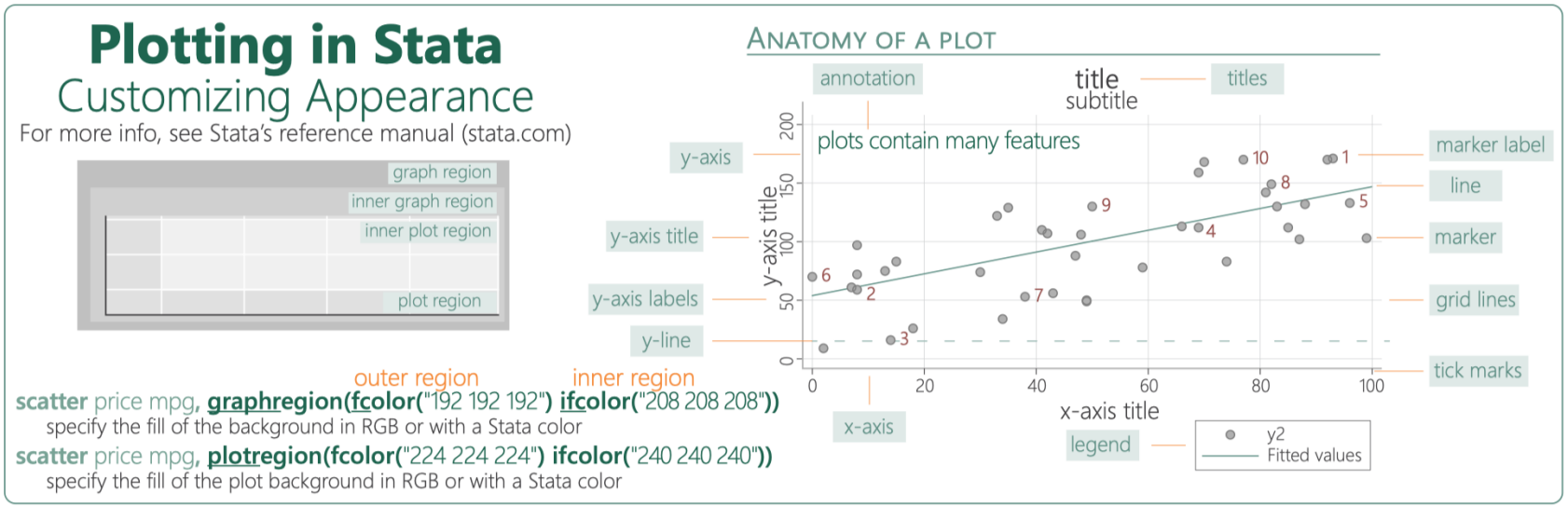
Plot Placement

Customizing Appearance

Save and Export Graph

Customizing Appearance

- The major reason to customize appearance is for **readability** and **ascetics**.
- For readability, we can give proper titles, legends, notes, and axis labels.
- For ascetics, we can change the color palette, the type and size for lines and markers...



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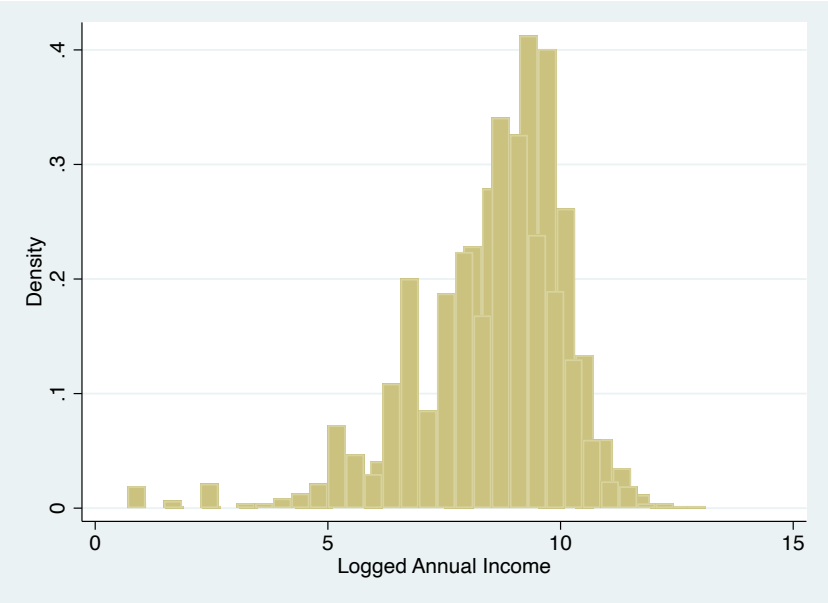
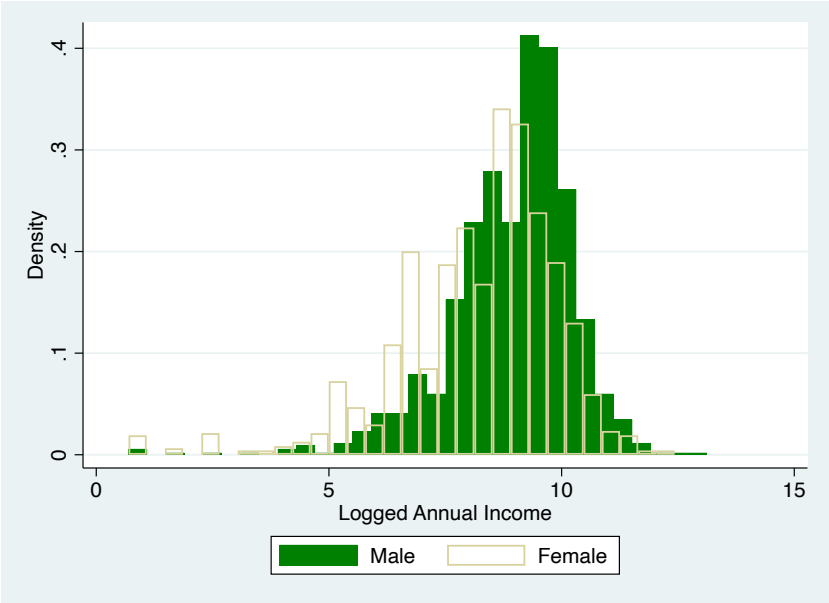
Customizing Color and Legends

```
. *** Appearance for readability and ascetic reasons
. twoway (hist loginc if gender==1, color(green)) ///
>         (hist loginc if gender==0, fcolor(none)), ///
>         legend(order(1 "Male" 2 "Female"))

. *help help twoway_options
```

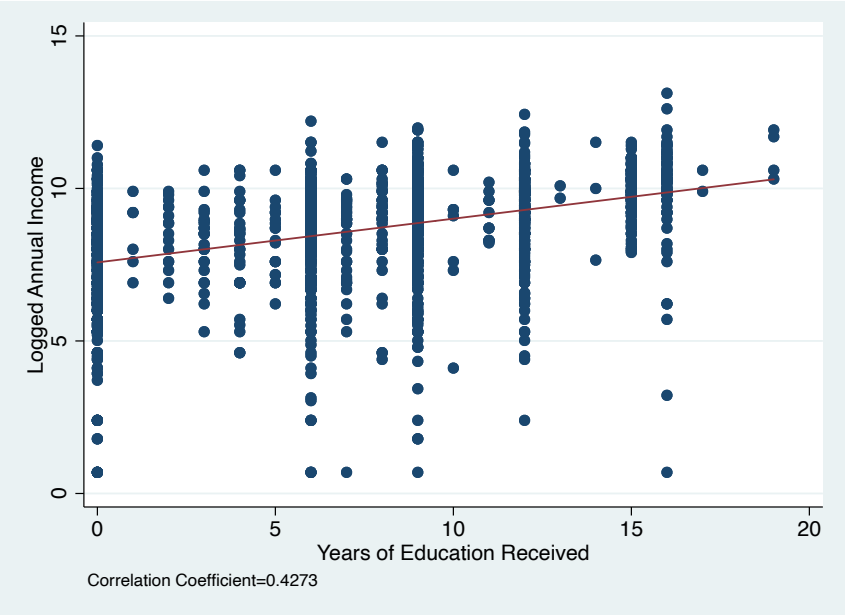
Customizing Color and Legends





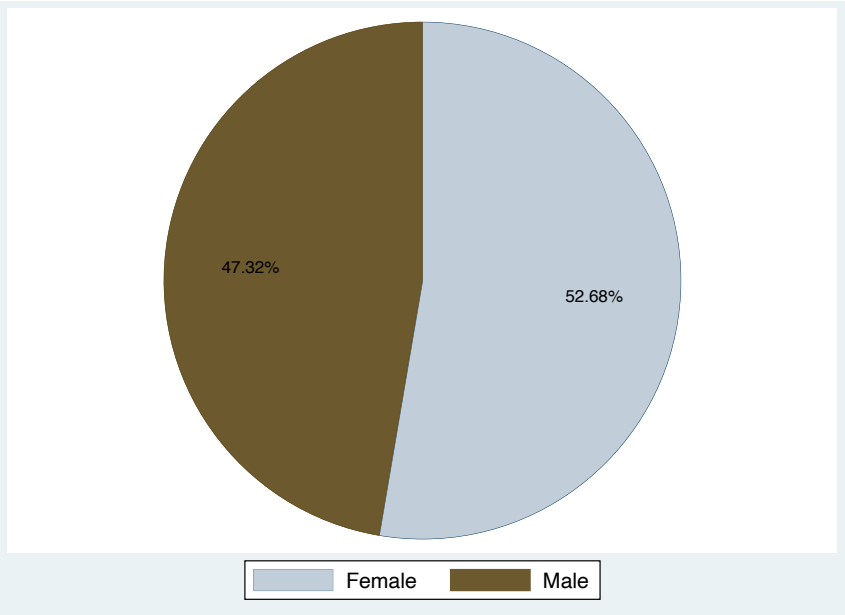
More Examples

```
. twoway (scatter loginc eduy) (lfit loginc eduy), ///
>         legend(off) ///
>         ytitle("Logged Annual Income") ///
>         note("Correlation Coefficient=0.4273")
```



More Examples

```
. #delimit ;
delimiter now ;
. graph pie, over(gender)
>         pie(1,color(navy%30))
>         pie(2, color(olive))
>         plabel(_all percent) ;
. #delimit cr
delimiter now cr
```



Scheme

Schemes are sets of graphical parameters, so you don't have to specify the look of the graphs every time.

Using a saved theme

```
twoway scatter mpg price, scheme(customTheme)
```

Create custom themes by saving options in a .scheme file

help scheme entries
see all options for setting scheme properties

adopath ++ "~/<location>/StataThemes"
set path of the folder (StataThemes) where custom .scheme files are saved

set as default scheme

set scheme customTheme, **permanently**
change the theme

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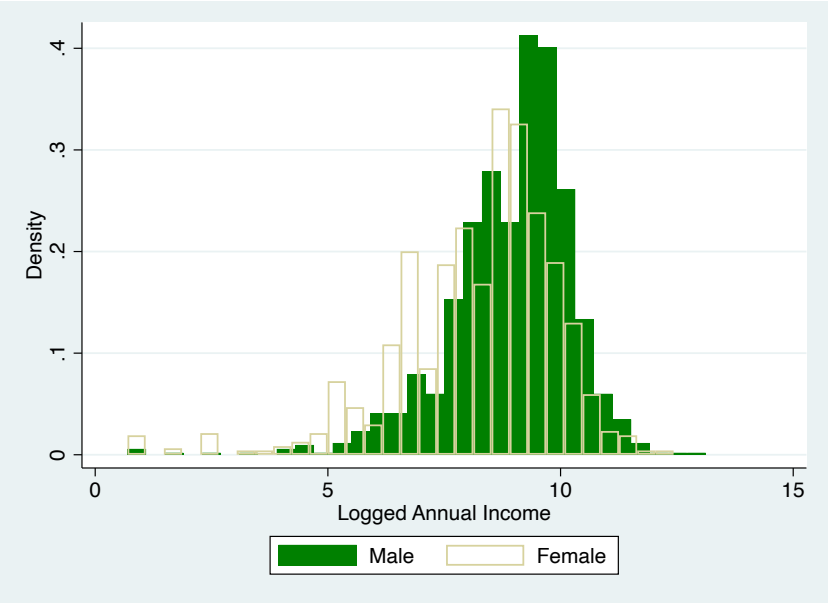
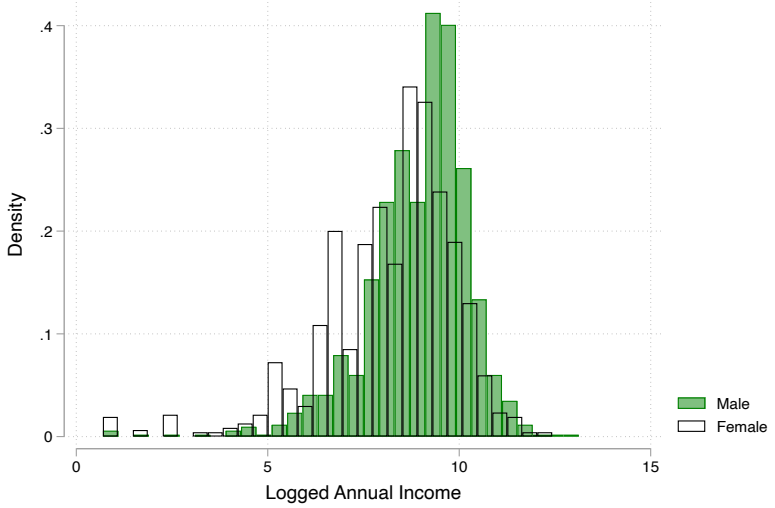
Brewscheme

- We can use the package "brewscheme"
- https://www.stata.com/meeting/switzerland16/slides/bischof-switzerland16.pdf

```
. *ssc install brewscheme
. graph box eduy, scheme(plotplain)
. graph box loginc, scheme(plottig)
. *you can also set your own schemes
.
.
. set scheme plotplain
. twoway (hist loginc if gender==1, color(green)) ///
>         (hist loginc if gender==0, fcolor(none)), ///
>         legend(order(1 "Male" 2 "Female"))
```

Brewscheme: plotplain





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Save and Export Graph

Save and Export Graph

```
. *** Save .gph file
. *graph save ...
.
. *** Export other plot types
. *graph export ...
```